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337312 (37)

BE (3rd Semester)

Examination, Nov-Dec., 2014

Branch : Mechanical

NUMERICAL ANALYSIS & COMPUTER PROG. (C & C++)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and 9d).

Q. 1. (a) Define absolute, relative and percentage error. **2**

(b) Find a real root of the equation $\cos x = 3x - 1$ correct to 4 decimal places by iteration method. **7**

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(c) Solve for a positive root of $x - \cos x = 0$ by Regula Falsi method. **7**

(d) Using Newton's method, find the root between 0 and 1 of $x^3 = 6x - 4$ correct to 5 decimal places. **7**

Q. 2. (a) Differentiate between curve fitting and an interpolation **2**

(b) From the data given below, find the number of students whose weight is between 60 and 70 **7**

Weight	40-45	45-50	50-60	60-70	70-80	80-90	90-100	100-110	110-120
No. of Students	250	120	100	70	50	30	20	10	5

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(c) State and derive Newton's backward interpolation formula. 7

(d) Apply Gauss's forward formula to obtain F(x) at x = 3.5 from the table below : 7

x :	2	3	4	5
f(x) :	2.626	3.454	4.784	6.986

Q. 3. (a) Write the Simpson's $\frac{3}{8}$th Rule. 2

(b) By dividing the range into ten equal parts, evaluate $\int_0^{\pi} \sin x dx$ by Trapezoidal and Simpson's rule. 7

(c) Using Taylor series method, find y at x = 0.1, 0.2 correct to three significant digits given : 7

$$\frac{dy}{dx} - 2y = 3e^x; \quad y(0) = 0$$

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(d) Using modified Euler method; find y(0.2), y(0.1) given : 7

$$\frac{dy}{dx} = x^2 + y^2, \quad y(0) = 1$$

Q. 4. (a) Classify the equation : 2

$$\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$$

(b) Solve $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial t} = 0$ given 7

u(0, t) = 0, u(4, t) = 0, u(x, 0) = x(4 – x)

Assume h = 1. Find the values of u upto t = 5.

(c) Solve $\nabla^2 u = \delta x^2 y^2$ for square mesh given u = 0 on the 4 boundaries dividing the square into 16 sub-squares of length 1 unit. 7

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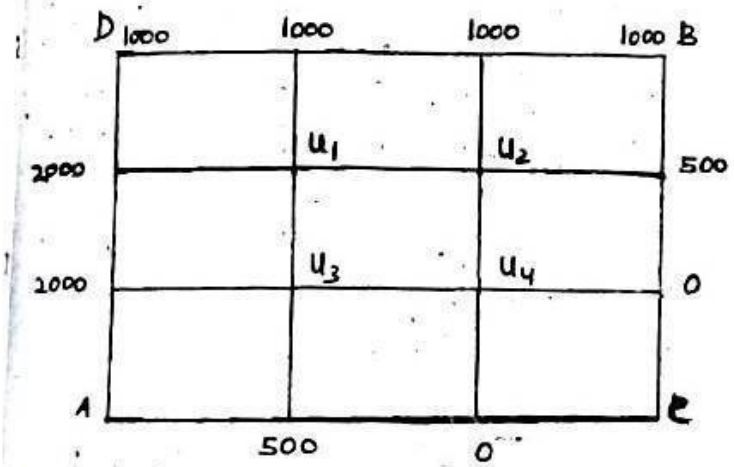
(d) Evaluate the function u(x, y) satisfying $\nabla^2 u = 0$ at the lattice points given the boundary values as follows : 7



2. 5. (a) Write the name of four basic types of constants in C ? 2

(b) What is Array ? How does it differ from ordinary variable ? 7

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(c) Explain relational, algebric & logical statements. 7

(d) Write program to find the factorial of a given number. 7

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